

Hricha Mehra

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| 📍 Mumbai, India

🔗 GitHub | in LinkedIn | </> LeetCode

Education

B.E. Computer Engineering — Mumbai University (CGPA: 8.91/10)

2022–2026

HSC — CBSE (80.4%)

2022

SSC — ICSE (95.6%)

2020

Experience

Software Engineer — HomeFirst Finance Company India Ltd.

Jan 2026 – Present

1. Test Data Generator

- Engineered an Apex-based test-data automation framework generating dependency-aware datasets across 20+ objects for streamlined test-data generation and execution.
- Designed an LLM powered layer to convert testing requirement into executable data generation configuration.

2. Security Control Center

- Designed a centralized access control auditing framework that performs periodic org-wide scans
- Leveraged dynamic SOQL on metadata to accurately resolve effective user permissions, ensuring access visibility

3. Loan Origination System

- Designed an in-house Loan Origination System with automated amortization schedule generation, eliminating dependency on external Loan Management Systems.
- Developed **Logistic Regression–based credit risk model** to predict Probability of Default, including data preprocessing, feature engineering and model evaluation for underwriting decisions. Integrated the model into the workflow.
- Built a secure Aadhaar data masking pipeline for sensitive customer information, ensuring regulatory compliance while preserving data usability for downstream processing.

Web development Intern — Chrysalis Technosoft

Mar 2025 – Jun 2025

- Built backend services for Job247 with JWT-based authentication, email verification and role-based authorization.
- Implemented Socket.IO real-time channels to enable instant recruiter–candidate updates and reduced latency.

Technical Skills

- **ML:** YOLO, CNNs, Grad-CAM, OpenCV, Prophet
- **Languages:** C++, Python, JavaScript
- **Backend:** Node.js, Express.js, Spring Boot, Socket.IO, MongoDB, MySQL
- **Tools:** Git, Postman, Vercel, Render
- **Frontend:** React, HTML, CSS
- **Core CS:** DSA, OOP, DBMS
- **Cloud:** AWS, Docker, Salesforce

Projects

AgroLens — AI-Powered Agriculture System

YOLOv8, Faster R-CNN, Grad-CAM

- Developed a weed detection pipeline using the Weed25 dataset (9,000+ field images across 25 weed classes) a 80:10:10 train-validation-test split for model training and evaluation.
- Performed hyperparameter tuning and benchmarked YOLOv8n, YOLOv8m, and Faster R-CNN on different parameters.
- Selected YOLOv8n for deployment after achieving 91.1% mAP@0.5, 88.0% F1-score, 87.8% Precision, 88.3% Recall, and real-time inference at 4 ms/image. Integrated Grad-CAM explainability to visualize model attention regions.
- Built a FastAPI inference service exposing REST APIs for image upload and real-time predictions, and integrated the model with a mobile application to deliver weed detection results with bounding boxes and confidence scores.

FinScope — Real-Time Finance Engine + Forecasting

Spring Boot, React, MySQL, Prophet

- Built secure REST APIs for budget tracking, and financial goal management with JWT-based user data isolation.
- Integrated Prophet-based forecasting pipelines trained on historical spending patterns to predict future expenses.
- Developed analytics dashboards visualizing category-wise spending, financial trends, and live balance updates for personalized financial insights.

FreshPress — Multi-Source News Aggregator

React, Express, RSS Parser, SendGrid

- Built a robust RSS ingestion pipeline aggregating TOI, Indian Express, LiveMint, The Hindu with less than 60s latency.
- Integrated SendGrid email digests to deliver daily news summaries and delivered a feature-complete UI with source filtering, preview summary & mobile responsiveness.

Achievements

- 1st Place — Multicon 2026 Paper Presentation
- 1st Runner-Up — HerSpark Ideathon (Aether 2025)
- 2nd Place — Multicon 2023 C/Cpp workshop
- Webmaster — CSI-TCET
- Documentation Team — TCET Shastra
- LeetCode: 550+ problems solved
- Backend team — TCET Open Source

Certifications

- ML Specialization — DeepLearning.AI (Andrew Ng)
- Full-Stack Development — Angela Yu
- AWS Cloud Practitioner Essentials
- Meta Front-End Developer Certification